

## Digital Workflow for Manufacturing Customizable Lower-Limb Orthosis

### Problem statements:

- Traditional lower limb orthosis fabrication is manual intensive, with long lead time and much material wastage.
- Digitalization of the subject can improve shape capturing efficiency and open up more possibilities in fabrication methods such as CNC mould making or 3D printed orthosis
- Market lacks high speed shape-capturing technology, especially for lower limb 3D scanning.

### Objectives:

- Develop an efficient shape-capturing technology.
- Develop a landmark recognition software plug in.

### Opportunities and value creation:

- Singapore fabricated more than 5000 orthosis annually
- Orthosis market is at 4.84b USD (2022)
- An efficient digital workflow increases demand for AM.

## Achievement

### Multi-lens 3D surface scanner

- Prototype (TRL5) completed.
- reduces patient compliance time to less than 10 seconds while maintaining scan accuracy comparable to existing solutions.
- Output of 3D mesh of the patient's leg within 5minutes.

### Trial in KKH

- 28 healthy adult volunteers participated in a proof-of-concept study. 56 limbs scanned within 2 hours. (Traditional plaster method takes 10-30minutes per limb)
- Users feedbacked that the overall process was surprisingly fast, device felt safe to use and easy for pediatric patients to comply without fear.

### Machine-learning-based landmark identification

- Plug-in that automatically detects key skeletal landmarks from the 3D mesh.
- Reduce clinician workload and enable more efficient orthosis design in existing software.

